Radar-based Sentiment Analysis and Recognition

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Introduction

Radar sensing for privacy-preserving affective computing

Sentiment analysis has traditionally relied on text, speech, and visual data, leaving body movement as a comparatively unexplored modality despite its strong link to affective expression. Radar sensing offers a non-intrusive, privacy-preserving alternative capable of capturing subtle body movements without the limitations of cameras or wearables.

This work uses the multimodal RF-Behavior dataset to investigate radar-based sentiment recognition through three complementary approaches:

- **Method 1:** Classical machine learning on kinematic, behavioral, and spatial features extracted from radar signals.
- **Method 2:** A transformer-based architecture trained end-to-end on raw radar and infrared sensor data.
- **Method 3:** A neural network framework that jointly performs radar signal interpolation and sentiment classification.

Dataset

The study is built on the third campaign of the RF-Behavior dataset [1], collected from 44 participants using 13 radars, 8 RFID tags, LoRa devices, and infrared cameras. This campaign targets six sentiment states: depression, distraction, excitement, focus, relaxation, and stress. Radar data consists of XYZ target-position sequences over time; infrared data adds 3–6 body-part landmarks (position and orientation) sampled at 100 Hz. Prior to modeling, radar streams are time-synchronized across sensors and normalized per user to remove individual biases.

Method 1: Feature-Based Classification

Radar trajectories were segmented into 90-frame windows with a 20-frame stride, then reduced to a set of kinematic, statistical, and directional descriptors (speed, acceleration, jerk statistics, spectral measures, tortuosity, turning-angle entropy, PCA-based directionality, etc.), summarized in Table 1.

Category	Examples
Motion dynamics	Speed, acceleration, jerk (mean, std, median, quartiles)
Statistical	Speed CV, skewness, autocorrelation, spectral entropy
Directional	Turn angle, tortuosity, PCA explained variance

Table 1: Extracted trajectory features (27 per radar).

With 13 radars, this yields 351 candidate features across over 600 windows. An ANOVA F-test selects the top-k features per radar, and Leave-One-Subject-Out (LOSO) cross-validation guides a joint search over window size, stride, and feature count for five classifiers: SVM, LightGBM, XGBoost, CatBoost, and MLP.

Results

The Support Vector Machine achieved the strongest performance with a mean weighted F1-score of 0.72, outperforming the gradient-boosting models (LightGBM and XGBoost around 0.66–0.67) and CatBoost (0.62). The MLP performed comparably to the boosting methods. These results suggest that carefully engineered motion descriptors can provide strong discriminative power without requiring deep architectures.

Method 2: Deep Learning with Raw Data

This approach trains transformer encoders directly on raw radar and infrared streams from all three RF-Behavior campaigns, avoiding manual feature engineering. Radar point clouds are resampled to a data-driven optimal rate of 18.7 FPS, balancing temporal frame count against spatial point density per frame.

The architecture (Figure 1) uses a Point Cloud Encoder (a BERT-style self-attention block, without positional encoding for radar) to condense spatial points into a single per-frame vector, followed by a Temporal Encoder (a LLaMA-style causal decoder with rotary embeddings) that condenses the frame sequence into a single recording-level embedding. For infrared data, an optional convolutional downsampling block can further compress the temporal axis.

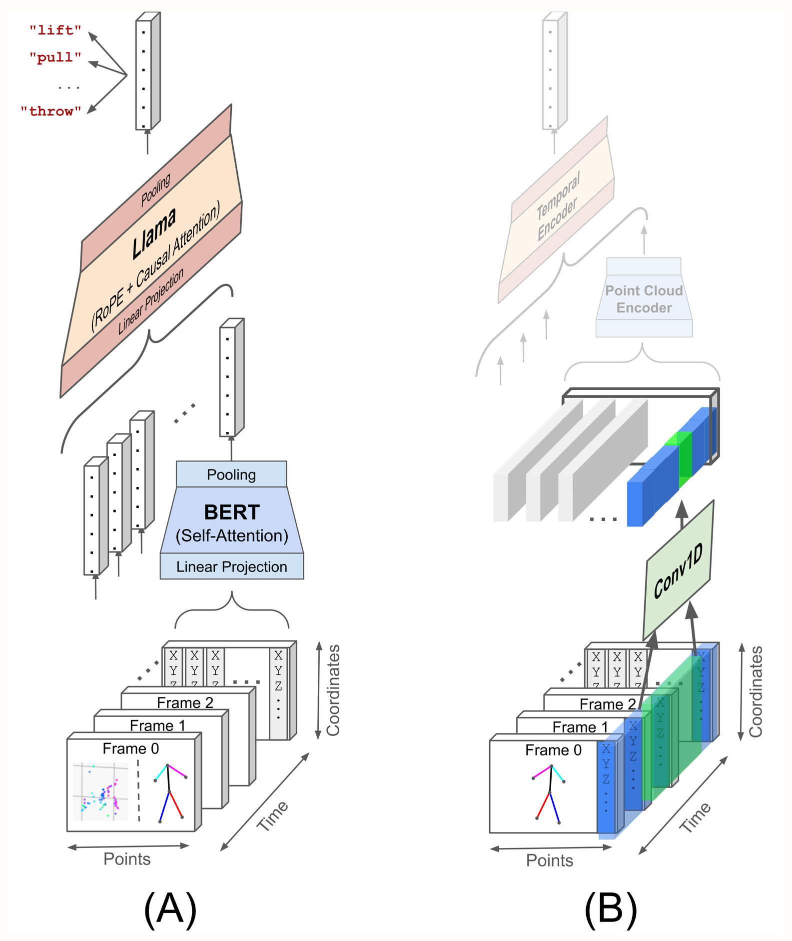

Figure 1: Transformer-based pipeline: a spatial Point Cloud Encoder processes 3D points per frame, followed by a temporal encoder over the resulting sequence of latent vectors.

Both supervised classification (cross-entropy loss, LOSO-style user split) and unsupervised contrastive learning (SimCLR-style NT-Xent loss, with CLIP-style radar/infrared alignment) were evaluated, along with a hyperparameter search over convolutional depth, transformer depth, attention heads, embedding size, and pooling strategy.

Results

Campaign	Classes	Type	Params	Test F1	Models
C1	21	Radar	1,311,125	94.41%	12
C1	21	IR	526,869	96.56%	25
C2	10	Radar	106,826	91.97%	23
C2	10	IR	1,541,850	98.12%	29
C3	6	Radar	1,349,766	46.12%	30
C3	6	IR	243,798	56.16%	56
C1+C2	31	Radar	120,895	89.51%	6
C1+C2	31	IR	195,375	91.52%	5
C1+C2+C3	37	Radar	268,453	83.74%	5
C1+C2+C3	37	IR	165,269	78.29%	7

Table 2: Performance metrics of supervised classification models.

Campaign	Split	Type	Params	Test F1	Models
C1	17-2-2	Radar	254,144	93.03%	6
C1	17-2-2	IR	142,016	96.68%	7
C2	6-2-2	Radar	93,440	77.51%	14
C2	6-2-2	IR	923,280	93.48%	31
C1+C2	23-4-4	Radar	175,776	90.31%	7
C1+C2	23-4-4	IR	282,176	88.13%	7
C1+C2+C3	25-6-6	Radar	817,472	79.20%	7
C1+C2+C3	25-6-6	IR	85,184	75.41%	7

Table 3: Grid search results for one-shot unsupervised classification. The split column indicates the number of train, validation, and test behavior classes.

Method 3: Neural Interpolation & Classification

Because the 13 radar sensors report at irregular, asynchronous timestamps, this method jointly learns to interpolate the sensor streams onto a common timeline and classify the resulting sequence into one of the six emotion classes.

Sequences are built per user/emotion by concatenating all 13 radars' XYZ values (39 features), with 10–30% of each sequence randomly withheld as interpolation targets. Three interpolation/classification models are compared against a cubic-spline-plus-ridge baseline: a Neural Controlled Differential Equation (NCDE) model, a 1D U-Net, and a Temporal Convolutional Network (TCN). The NCDE integrates a differentiable Hermite-spline control signal to obtain a continuously updating hidden state, in contrast to the discrete updates of RNNs.

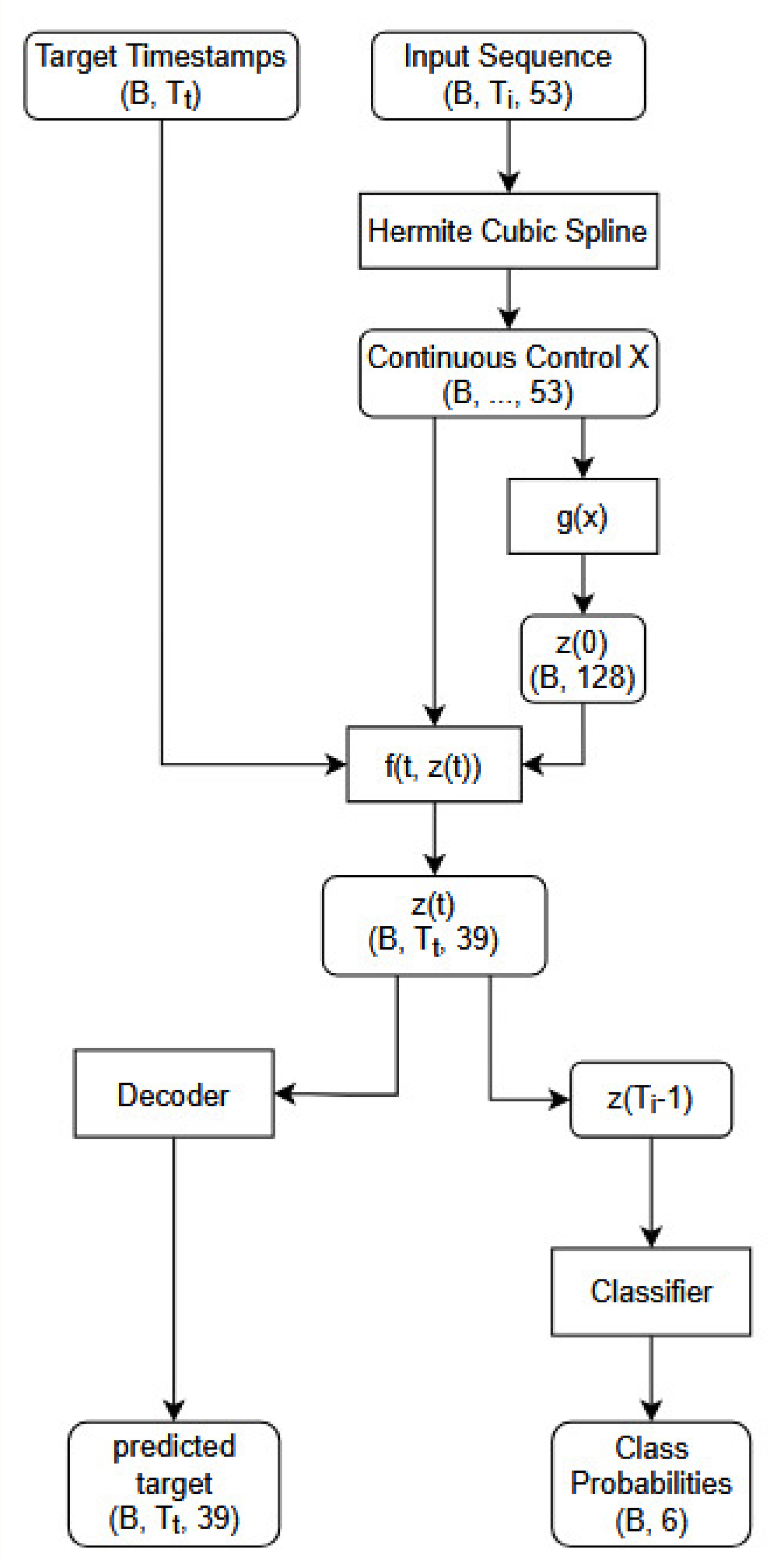

Figure 2: Workflow of the NCDE-based interpolation and classification pipeline: the sparse input sequence is converted to a continuous Hermite-spline control signal, integrated to obtain hidden states, then decoded for value reconstruction and classified for emotion probabilities.

Results

Method	MSE	Loss	Accuracy (%)
Baseline	0.0962	–	–
NCDE	0.0999	3.617	44.44
U-Net	0.0995	1.792	16.67
TCN	0.0992	1.744	25.00
U-Net + Baseline	0.0962	1.792	16.67
TCN + Baseline	0.1039	1.650	33.33

Table 4: Interpolation and classification results.

Summary and Conclusions

Across all three methodologies, radar sensing shows clear promise for privacy-preserving sentiment recognition, though performance on subtle emotional states remains behind that on simpler, more repetitive behaviors, and behind infrared-based approaches. Feature-based SVM classification proved the most robust and data-efficient approach on the emotion classes studied. Deep learning on raw data generalizes well to unseen behaviors in the unsupervised setting but degrades on fine-grained emotion recognition. Joint interpolation-classification remains constrained by limited data.

Promising future directions include multimodal fusion of radar with infrared, data augmentation (sensor dropout, noise injection, geometric transforms), architectures tailored to point-cloud and irregular-time-series structure, and aligning learned sensor representations with text encoders to enable zero-shot recognition of new behaviors.

References

[1] S. Zuo, Y. Song, S. Golipoor, Y. Liu, X. Ma, S. Sigg, RF-Behavior: A multimodal radio-frequency dataset for human behavior and emotion analysis, 2025.

[2] T. Chen, S. Kornblith, M. Norouzi, G. Hinton, A simple framework for contrastive learning of visual representations, *ICML*, 2020.

[3] A. Radford et al., Learning transferable visual models from natural language supervision, *ICML*, 2021.

[4] A. Vaswani et al., Attention is all you need, *NeurIPS*, 2017.

[5] P. Kidger, J. Morrill, J. Foster, T. Lyons, Neural controlled differential equations for irregular time series, *NeurIPS*, 2020.